and advanced visualizations, such as Honeycomb's BubbleUp (https://www.honeycomb.io/bubbleup).

Practically speaking, this is a very simplified view of the potential costs involved, and you should work with your vendor of choice to come to a better understanding of the costs in your specific case, including appropriate discounts, where applicable. While there *are* costs, don't forget that at the end of the day, the return you get from distributed tracing is that you're not flying blind anymore with your microservices app and can expect a decrease in mean time to repair/recover (MTTR) and an increase in developer productivity.

8.3.3 Traces vs. metrics vs. logs

Keep in mind that one signal type alone can't usually answer all the questions you may have to arrive at actionable insights. We will dive deep into that topic in chapter 11.

Independent of correlation, you can sometimes derive signals from each other. For example, you can create metrics based on observed spans, such as with OpenTelemetry's span metrics connector (http://mng.bz/Aoez). The distributed tracing space is evolving quickly, and various vendors and communities are experimenting and iterating (https://keyval.dev/distributed-tracing-2025/) to improve both its performance and usability, including automation of collection and compression of telemetry signals.

Summary

- Distributed tracing provides you with the *where* and *what* concerning visibility into distributed systems.
- An execution is distributed over microservices, and to get the overall picture, the various moving parts must be stitched together via context propagation.
- A trace is the end-to-end processing of a request in a distributed system, with a span representing a unit of work, such as the invocation of a single service or a call to a database.
- There is a long list of possible use cases for distributed tracing, from monitoring microservices-based applications to latency optimization to troubleshooting.
- There are several costs involved with distributed tracing, including instrumentation costs to transport and ingestion costs to query costs.
- You will have to work backward from your use case to determine the return on investment (see chapter 2 for a definition), making sure your stakeholders understand the value, be that MTTR reduction or, indeed, an increase in developer productivity.
- A signal type, such as metrics or logs, alone can't usually produce actionable insights; hence, support for signal correlation is a critical factor in selecting a distributed tracing solution.